

Component & Credentials List

AI Lead Qualification System — Capstone Project (Option 3)

Petalnex Pvt. Ltd. — AI Automation Internship
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Node Groups & Integrations

Component	Type	Purpose
Webhook Trigger	Trigger	Receives lead form POST request (name, email, company, phone, message)
Validation	Code / IF node	Checks required fields, validates email format, flags duplicates
Business Logic	Code node	Normalizes fields, adds timestamp, prepares structured prompt for AI Agent
AI Agent	LLM node	Qualifies lead intent and assigns a 0-100 score with reasoning
Supabase Insert/Update	Database node	Writes lead record; updates score/status matched by id (UUID)
IF — Branch on Score	Logic node	Routes lead to hot-alert path or nurture path based on score threshold
Gmail SMTP — Sales Alert	Notification node	Sends immediate email alert to sales team for hot leads
Gmail SMTP — Nurture	Notification node	Sends follow-up/nurture email sequence for warm/cold leads

AI Models & Data Stores

Service	Role	Purpose
Gemini (AI Studio)	Primary LLM	Lead qualification reasoning + scoring
Groq	Backup LLM	Fallback if Gemini free-tier quota is hit during testing
Supabase (Postgres)	Primary data store	`leads` table — stores lead details, score, status, timestamps

Credentials Required

- Gemini API key (Google AI Studio) — primary LLM access

- Groq API key — backup LLM access
- Supabase project URL + service role key — Transaction pooler, port 6543
- Gmail SMTP app password — sajeelali307@gmail.com, credential named "SMTP account"

Failure Handling

- Malformed or incomplete webhook payloads are rejected at the Validation step before reaching the AI or database layers
- LLM timeout or quota errors on Gemini trigger a fallback call to Groq
- Structured JSON output from the AI Agent is parsed and validated in a deterministic Code node rather than relying on autoFix retries
- Supabase writes are matched on the `id` (UUID) column to avoid overwriting the wrong record in multi-lead scenarios

Security Considerations

- Gmail SMTP credential is scoped to sending only — no inbox read access required for this workflow
- No lead PII is sent to third-party enrichment APIs unless explicitly scoped and approved
- Supabase table access is restricted to the service role used by n8n; RLS reviewed before granting broader access
- Webhook endpoint validates payload structure before any downstream processing to reduce injection risk